



**Escola Superior
de Tecnologia
e Gestão**

Politécnico de Coimbra

Inteligência Artificial

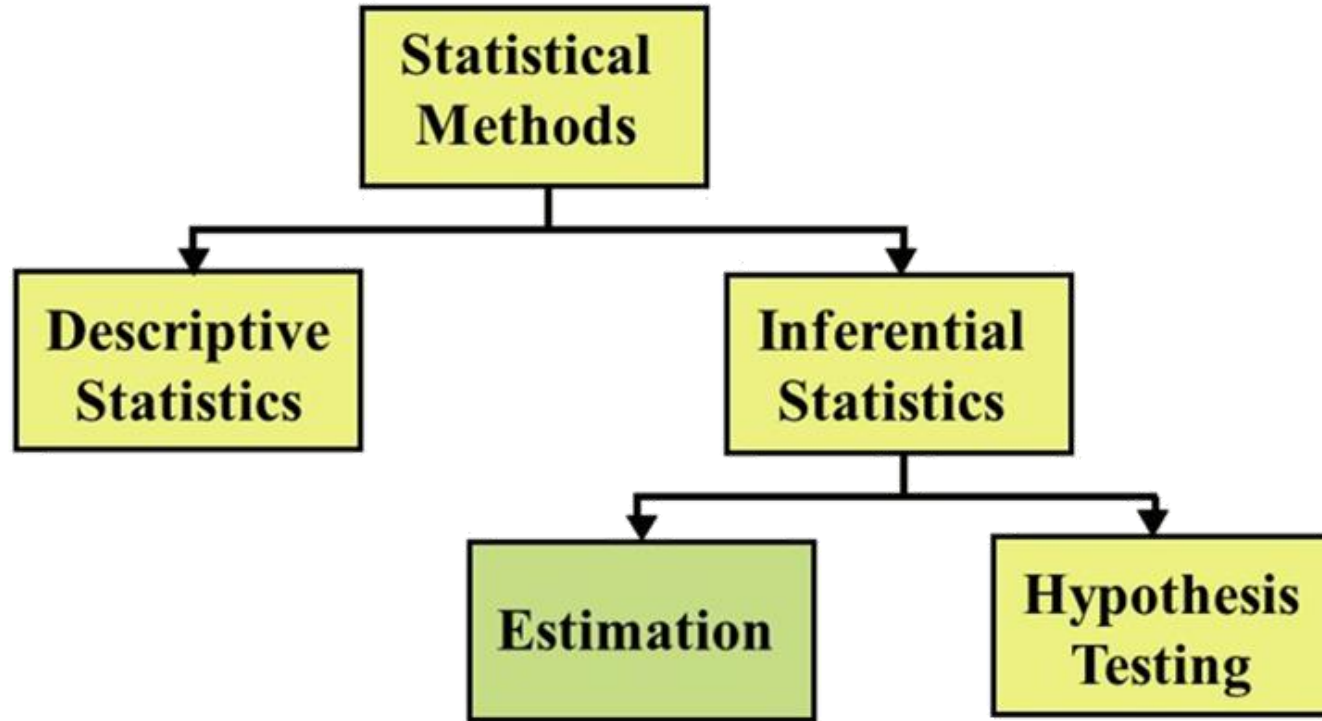
Parametric Tests

Table of Contents

- **Statistics**
- **Types of Tests**
- **Levene Test**
- **Parametric tests**
 - Requirements
 - **Student's t-test**
 - One Sample t-test
 - Unpaired Samples t-test
 - Paired Samples t-test
 - **ANOVA**
 - One-way ANOVA
 - Requirements
 - Post-hoc Tests
 - One-way ANOVA with repeated measurements
- **SPSS**

Inferential Statistics

- **Statistics**



<https://slideplayer.com/slide/4505649/>

Population:

- values such as means, standard deviations, and proportions when referring to a population are called ***parameters***.
- since it is usually difficult to gather data from an entire population, we cannot know the true values of the parameters for a population

Sample:

- the estimates that come from a sample are called ***statistics***.
- sample statistics are used to estimate population parameters.

Inferential Statistics

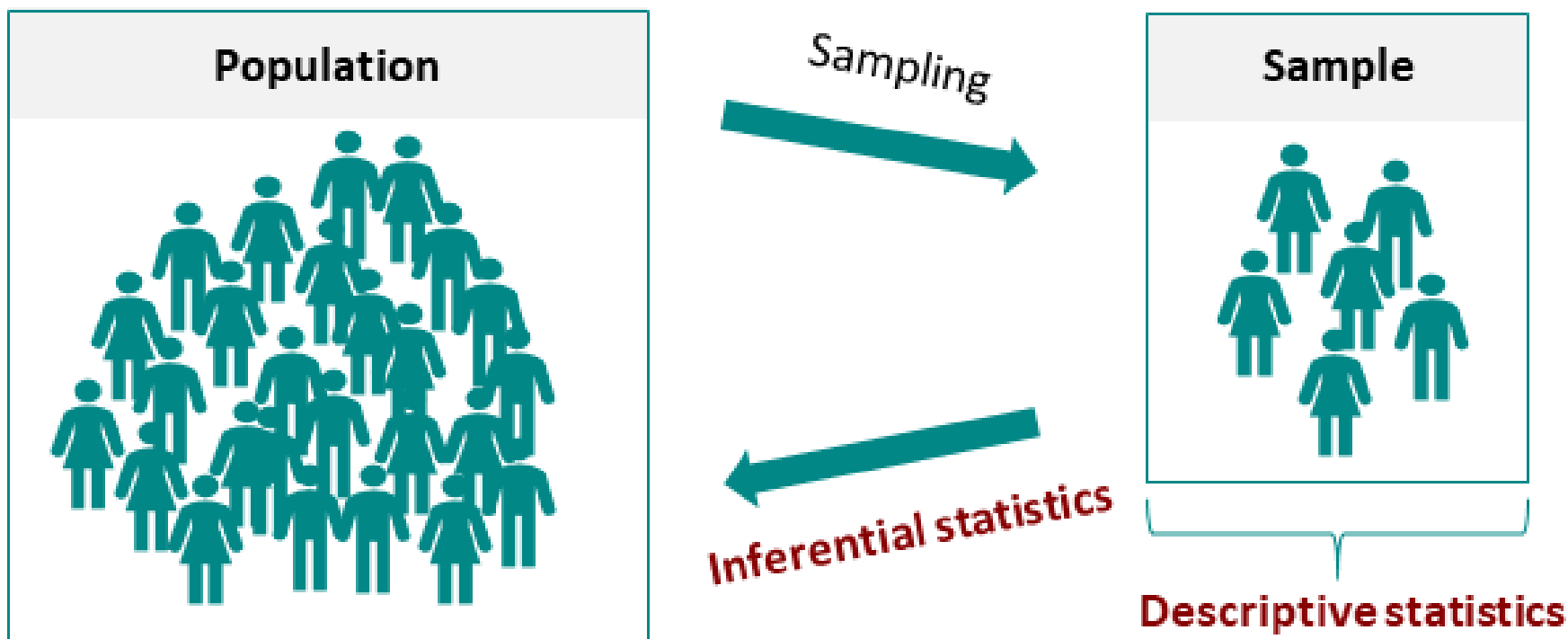
- **Statistics**

Descriptive statistics:

- provides tools to describe a sample.

Inferential statistics:

- based on a sample, makes a statement about the population.



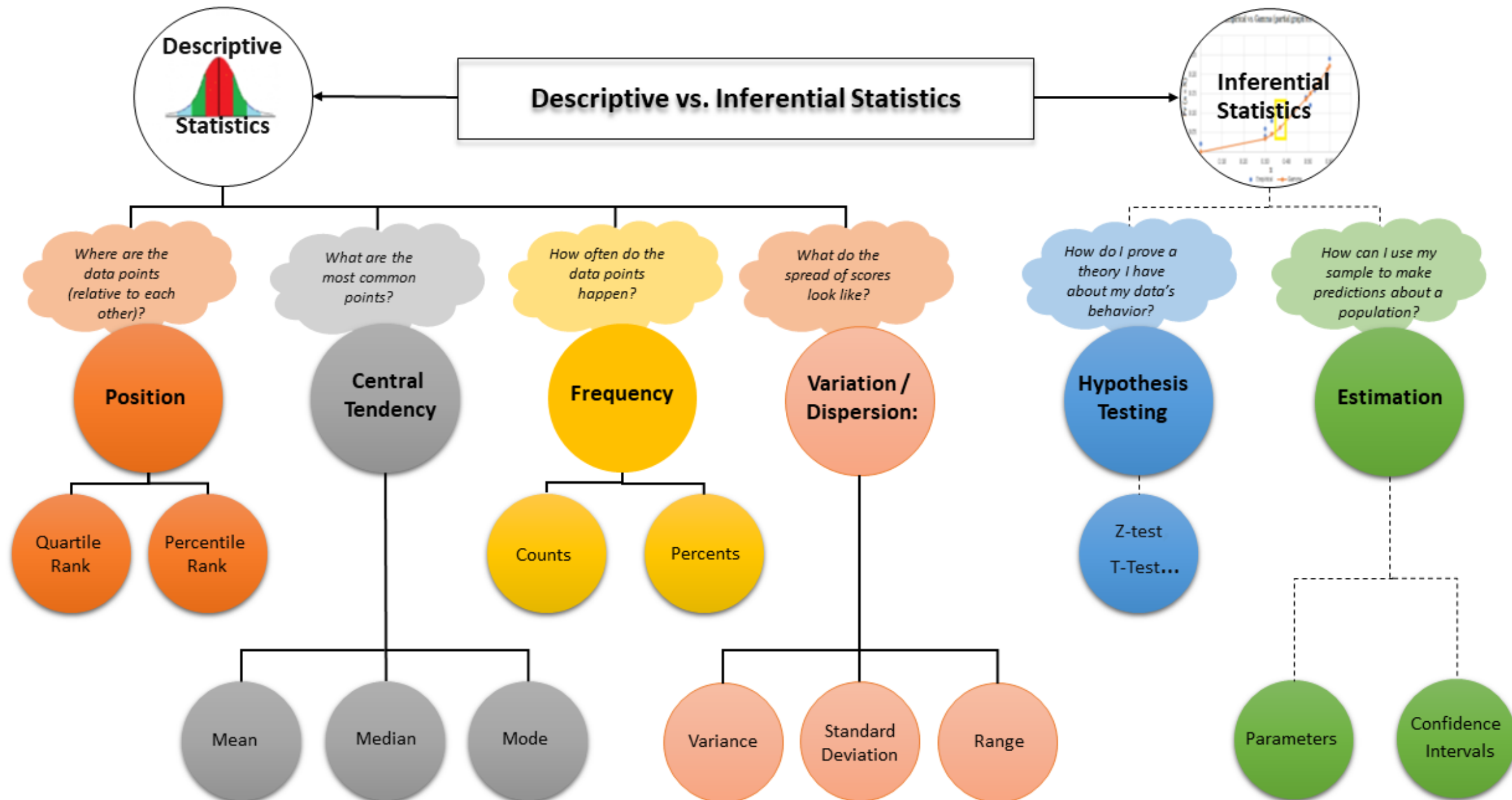
Inferential Statistics

- **Statistics**

DESCRIPTIVE	INFERENTIAL
It is the analysis of data that helps to describe, show and summarize data under study	It is the analysis of random sample of data taken from a population to describe and make inference about the population
Organize, analyze and present data in a meaningful way	Compares, test and predicts data
It is used to describe a situation	It is used to explain the chance of occurrence of an event
It explain already known data and limited to a sample or population having small size	It attempts to reach the conclusion about the population
Types: Measure of central tendency & Measure of variability	Types: Estimation of parameters & Testing of hypothesis
Results are shown with help of charts, graphs, tables etc.	Results are shown with help of probability scores

Inferential Statistics

<https://st11.ning.com/topology/rest/1.0/file/get/3445641805?profile=original>



Inferential Statistics

- **Statistics**

Estimation is using data from a sample to estimate or predict a parameter of interest in a population.

Hypothesis testing is using data from a sample to assess the validity of a statement about a population parameter.

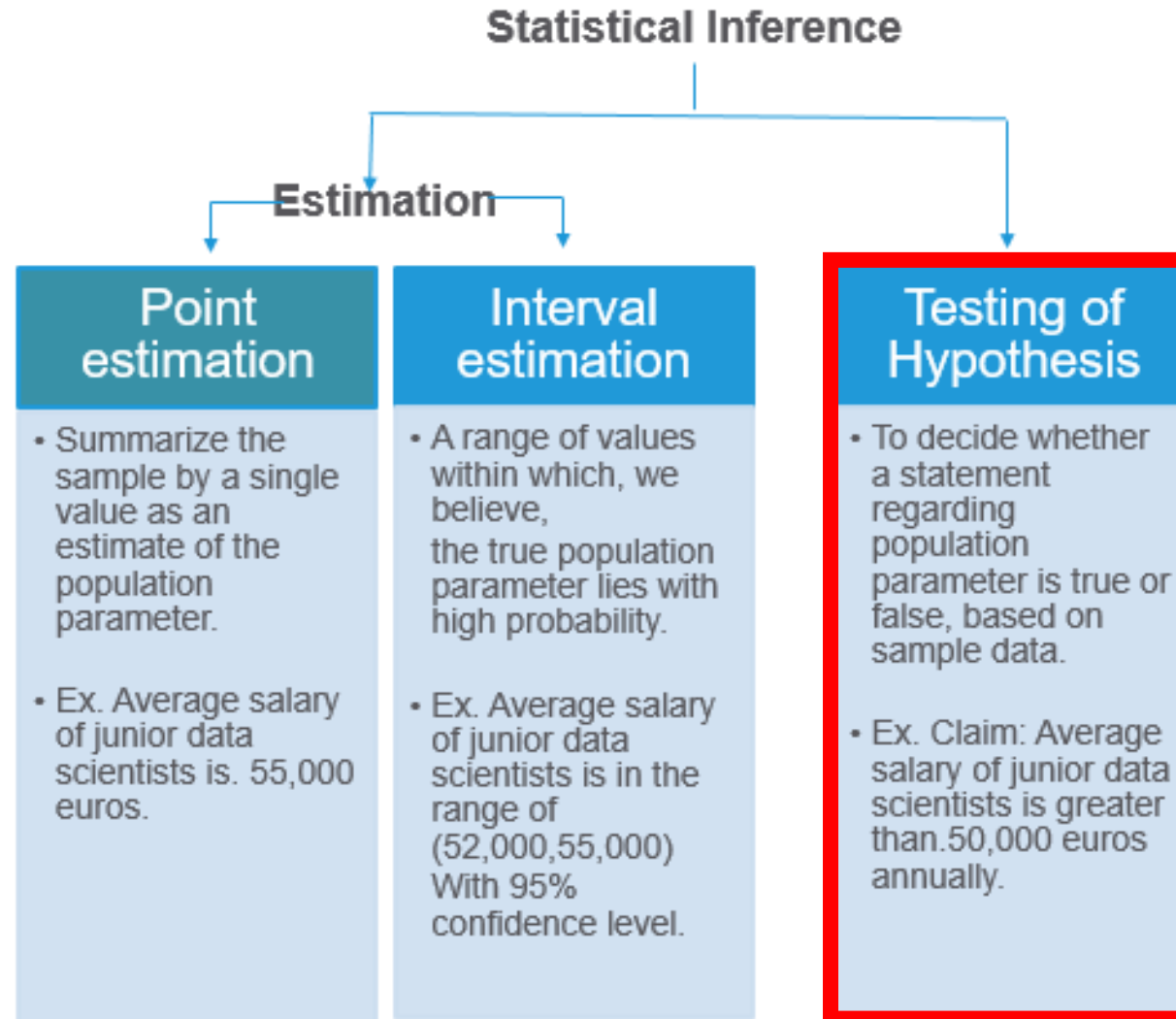


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Inferential Statistics

- **Types of Tests**

- **Parametric statistical tests**

- assumes certain things about the population parameters and the distributions that the data come from.
 - main assumption that these tests make is that the values are approximately **normally distributed** in the population

- **Nonparametric statistical tests**

- does not assume anything about the population parameters.
 - often called “distribution-free tests” since they do not assume an underlying distribution.

Inferential Statistics

- Types of Tests

	Parametric	Nonparametric
Describe one group	Mean SD; SEM; 95%CI	Median Range
Compare one group to a hypothetical value	1-sample t-test	Wilcoxon test (or Chi-square for two possible outcomes)
Compare two unpaired groups	2-sample unpaired t-test	Mann-Whitney test
Compare two paired groups	2-sample paired t-test	Wilcoxon test
Compare three or more unmatched groups	One-Way ANOVA	Kruskal-Wallis test ((or Chi-square for two possible outcomes)
<i>Predict value from measured values</i>	<i>Linear or nonlinear regression</i>	<i>Nonparametric regression</i>

<https://biocomm.eu/2019/01/10/guide-to-essential-biostatistics-in-the-scientific-method-3-2-2-2-2-2-2-2-2-2/>

Inferential Statistics

- **Types of Tests**

- **Reasons to Use Nonparametric Tests:**

- Nonparametric tests are valid when data does not meet the assumptions associated with the desired parametric test.
 - Nonparametric tests are often a good option for **small sample sizes** where parametric assumptions of normality are worrisome.
 - Nonparametric tests can be used when an area of study is best represented by the median (e.g. income)
 - Nonparametric tests can be used to analyse ordinal data, ranked data, or when outliers can't be removed from the data.

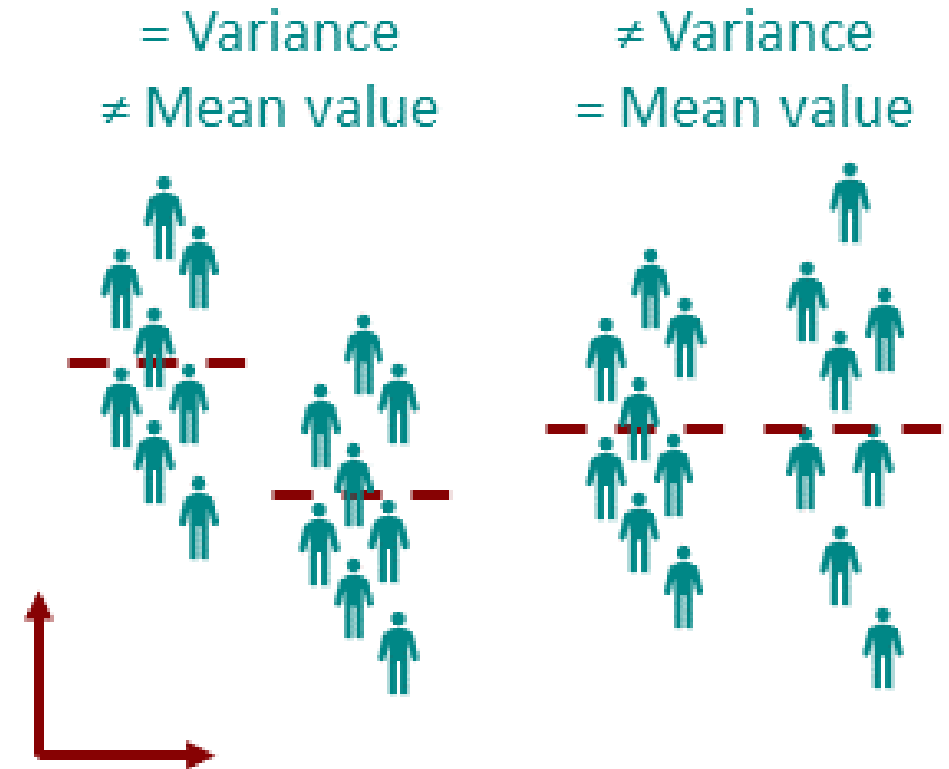
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Inferential Statistics

- **Levene Test**

- several statistical testing procedures require that there is equal variance in the samples.
- checks whether several groups have the same variance in the population
 - **H0: Groups have equal variances**
 - H1: Groups have different variances
- Assumptions for the Levene test:
 - independent observations
 - the test variable has metric scale level



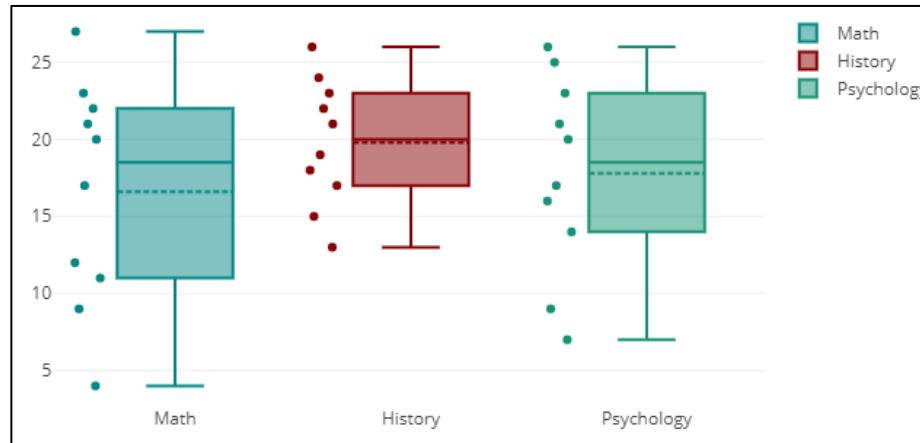
Inferential Statistics

- Levene Test

Cups of coffee per week

Math	History	Psychology
21	18	17
23	22	16
17	19	23
11	26	7
9	13	26
27	24	9
22	23	25
12	17	21
20	21	14
4	15	20

	N	Mean	Std. Deviation
Math	10	16.6	7.291
History	10	19.8	4.131
Psychology	10	17.8	6.443
Total	30	18.067	6.04



Level test of variance equality			
F	df1	df2	Significance
2.016	2	27	0.153

0.153 is greater than the 0,05, then the null hypothesis is maintained, and there is NO difference between the variances

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Inferential Statistics

- **Parametric tests**

	Parametric	Nonparametric
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Compare three or more unmatched groups	One-Way ANOVA	Kruskal-Wallis test ((or Chi-square for two possible outcomes)
<i>Predict value from measured values</i>	<i>Linear or nonlinear regression</i>	<i>Nonparametric regression</i>

Inferential Statistics

- **Parametric tests**
 - **Requirements**
 - **Os testes paramétricos assumem:**
 - a existência de distribuição normal
 - variáveis quantitativas, ou ordinais com escala de 7 ou mais níveis
 - **Para que a distribuição normal possa ser assumida basta que UM dos seguintes requisitos seja satisfeito:**
 - **Testes de Normalidade:**
 - > 50 casos -> **Kolmogorov–Smirnov (KS)**
 - < 50 casos -> **Shapiro–Wilk (SW)**
 - **Medidas de Assimetria:**
 - Coeficiente de assimetria (**skewness**)
 - Coeficiente de achatamento (**kurtosis**)
 - **Teorema do Limite Central (TLC):**
 - Mínimo de 30 casos

H0 = distribuição normal:
(sig, p-value, p) > 0,05 -> aceitar H0 -> distribuição normal
Analyse -> Descriptive -> Explore -> Plots -> Normality plots with tests

Distribuição simétrica => $-1,96 < \text{"coeficiente skewness"} / \text{"erro padrão"} < 1,96$
Considera-se assimetria tolerável se valor no intervalo $[-3; 3]$
Analyse -> Descriptive -> Freq -> Stat

Distribuição Mesocúrtica => $\text{"coeficiente kurtosis"} / \text{"erro padrão"} \approx 3$
Analyse -> Descriptive -> Freq -> Stat

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Inferential Statistics

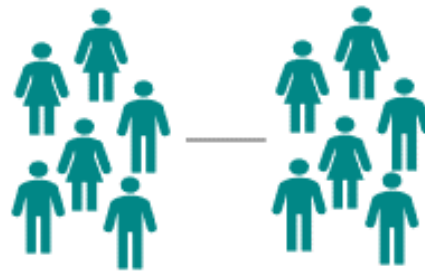
- Parametric tests
- Student's t-test
 - determines either whether the **group mean** and the mean of the population, or two group means differ statistically:

One sample t-test



Is there a **difference** between a **group** and the **population**

(Unpaired) Independent samples t-test



Is there a **difference** between **two groups**

Paired samples t-test



Is there a **difference** in a **group** between **two points in time**

Inferential Statistics

- Parametric tests
- Student's t-test
 - One Sample t-test
 - examines whether the mean of a population is statistically different from a known or hypothesized value

Two-tailed (non-directional)

- **Null hypothesis H_0 :** The mean value of the population is equal to the specified value.
- **Alternative hypothesis H_1 :** The mean value of the population is not equal to the specified value.

One-tailed (directional)

- **Null hypothesis H_0 :** The mean value of the population is equal to or greater than (or less than) that of the specified value.
- **Alternative hypothesis H_1 :** The mean value of the population is smaller (or larger) than the specified values.

Inferential Statistics

- Parametric tests
- Student's t-test
 - Unpaired Samples t-test
 - whether there is a difference between **two** unrelated groups

Null hypothesis H_0

There is no mean difference between the two groups in the population

→ Two population means are equal.

→ The two groups are from the same population

→ $H_0: \mu_1 = \mu_2$

Example: There is **no difference** between the salary of men and women

Alternative hypothesis H_1

There is a mean difference between the two groups in the population

→ The two population means are not equal

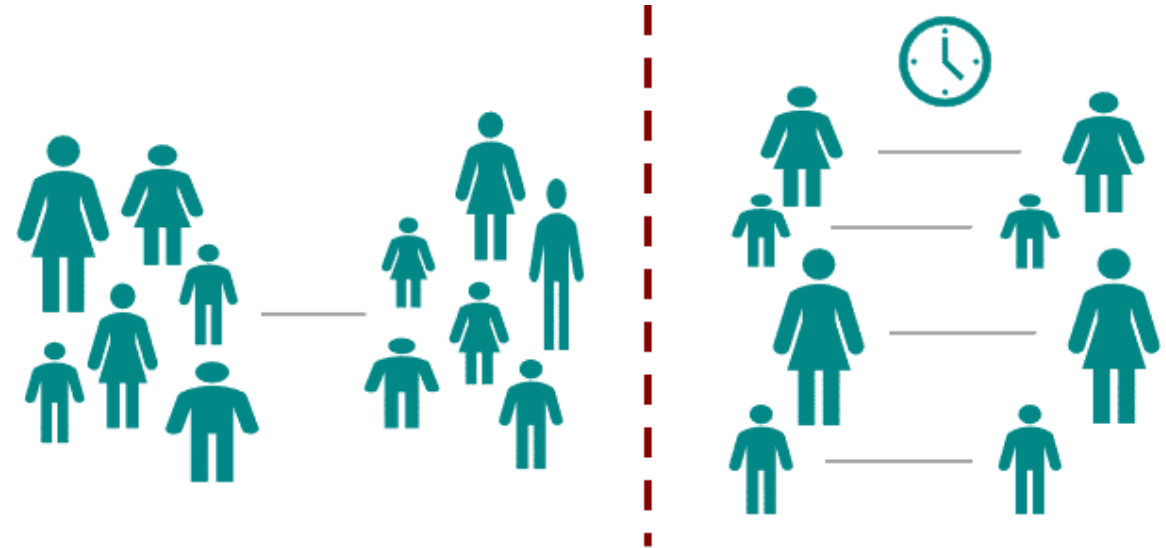
→ The two groups are not from the same population

→ $H_1: \mu_1 \neq \mu_2$

Example: There is **a difference** between the salary of men and women

Inferential Statistics

- Parametric tests
- Student's t-test
 - Paired Samples t-test
 - determines whether there is a difference between two dependent groups
 - needed whenever we survey the same group or sample at two points in time.



- **Null hypothesis** H_0 : The mean value of the two dependent groups is equal.
- **Alternative hypothesis** H_1 : The mean values of the two dependent groups are different.

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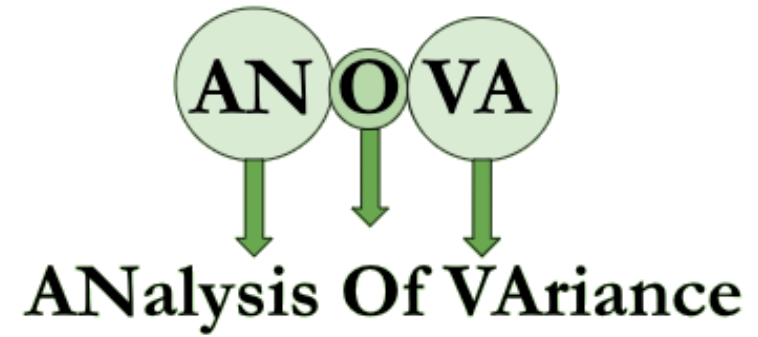
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- **SPSS**

Inferential Statistics

- Parametric tests

- ANOVA

- checks whether there are statistically significant differences between more than two groups
 - in contrast to the t-test, which can compare only two groups, the ANOVA compares more than two groups.
- the mean values and variances of the respective groups are compared with one another
- Variables:
 - **Independent:** nominal (with at least two values), or ordinal with high amplitude
 - **Dependent:** quantitative

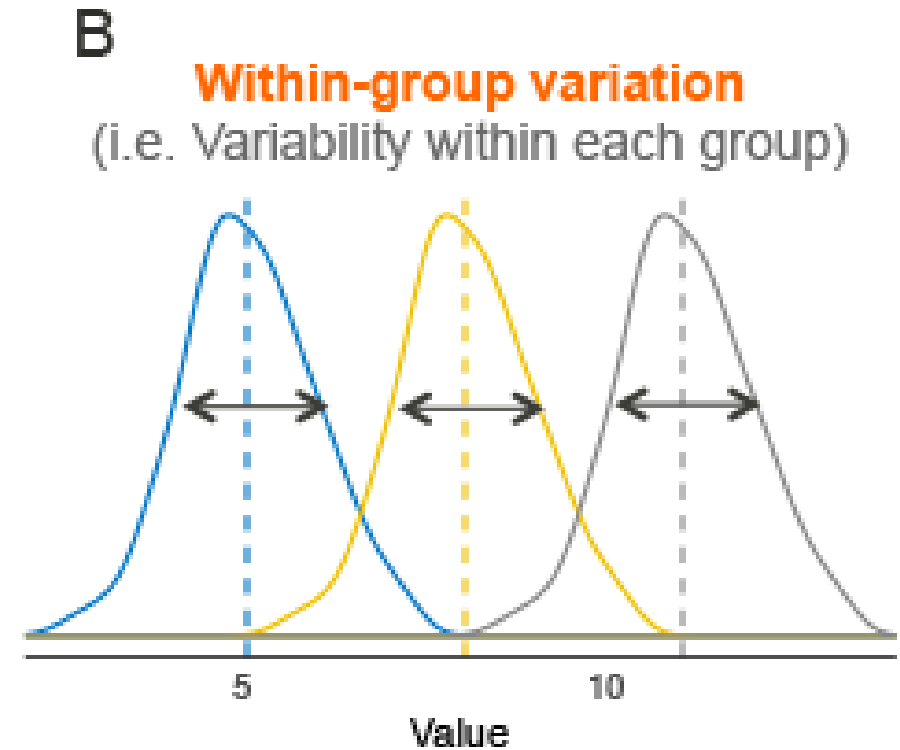
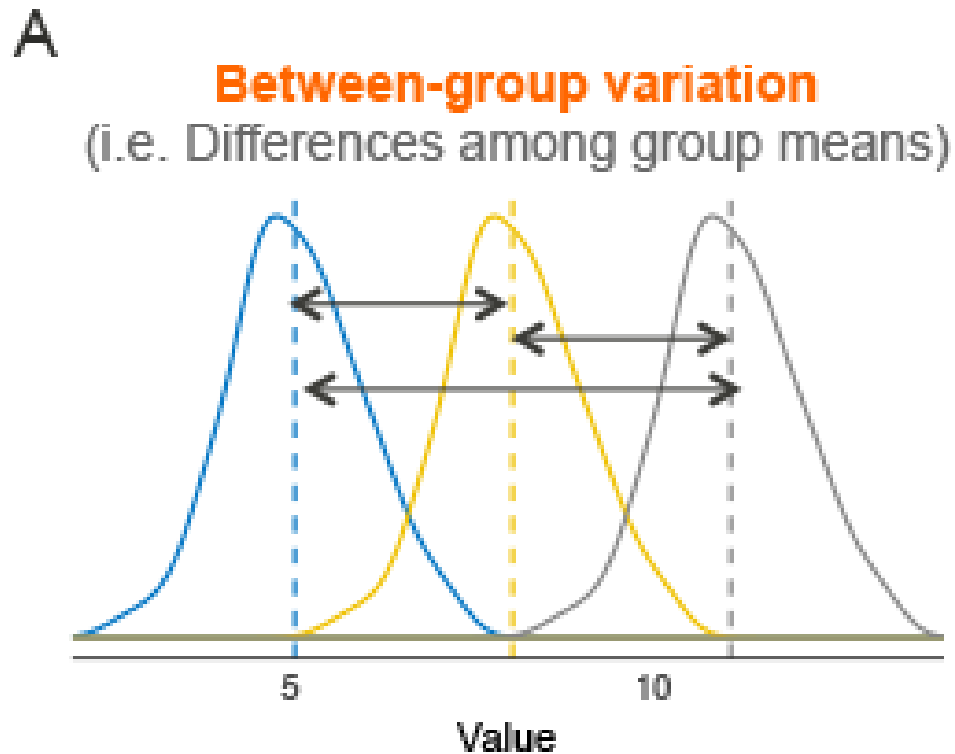


Inferential Statistics

- Parametric tests
- ANOVA

The idea behind the **ANOVA** test is very simple:

- if the average variation between groups is large enough compared to the average variation within groups, then at least one group mean is not equal to the others.
- Therefore, it's possible to evaluate whether the differences between the group means are significant by comparing the two variance estimates.
 - This is why the method is called analysis of variance even though the main goal is to compare the group means.



Inferential Statistics

- Parametric tests

- ANOVA

- Types:

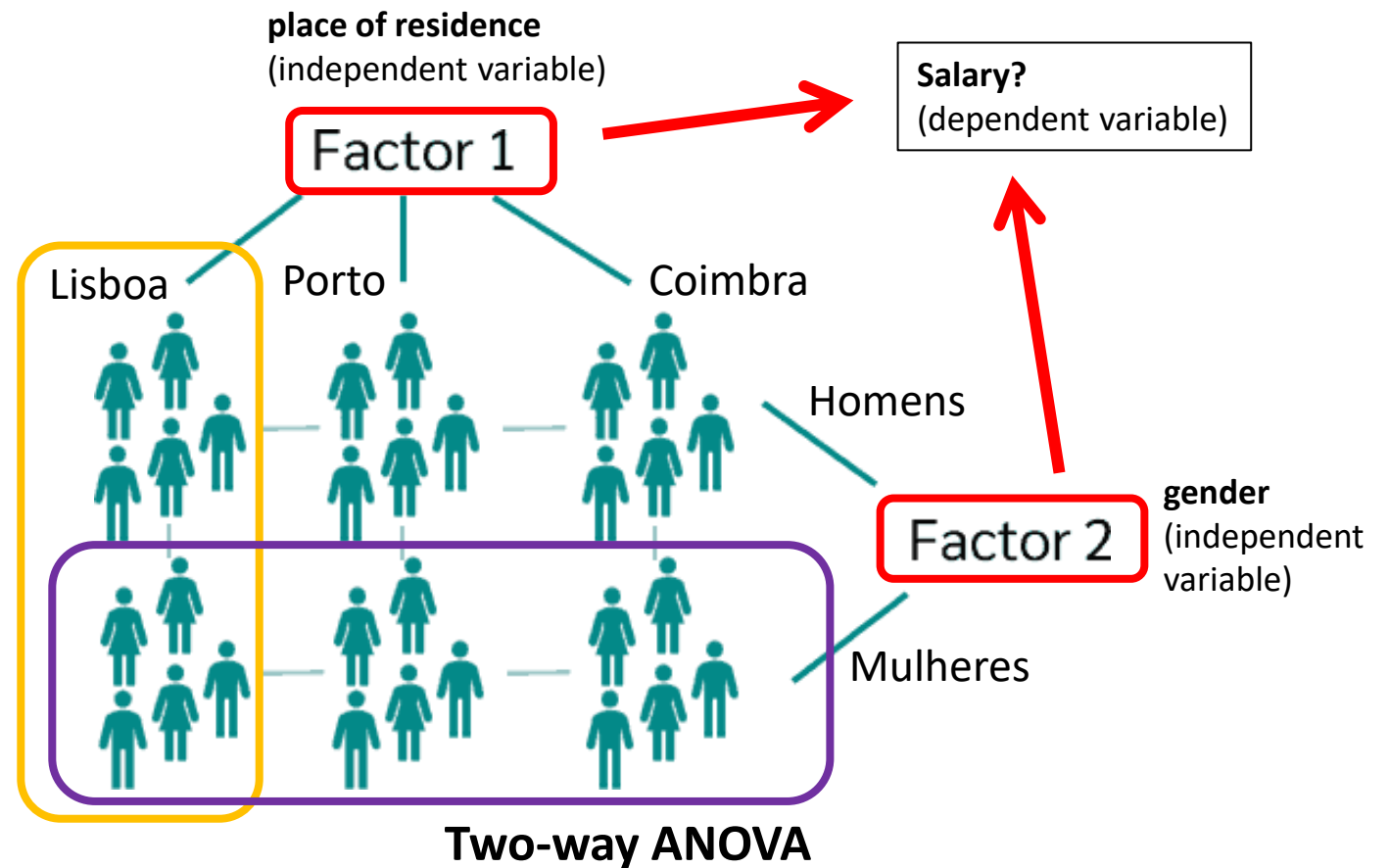
- One-way ANOVA

- whether one independent variable has an influence on a metric dependent variable.

- Two-way ANOVA

- whether two independent variables have an influence on a metric dependent variable.

One-way ANOVA	Two-way ANOVA
Does a person's place of residence (independent variable) influence his or her salary?	Does a person's place of residence (1st independent variable) and gender (2nd independent variable) affect his or her salary?



Inferential Statistics

- Parametric tests

- ANOVA

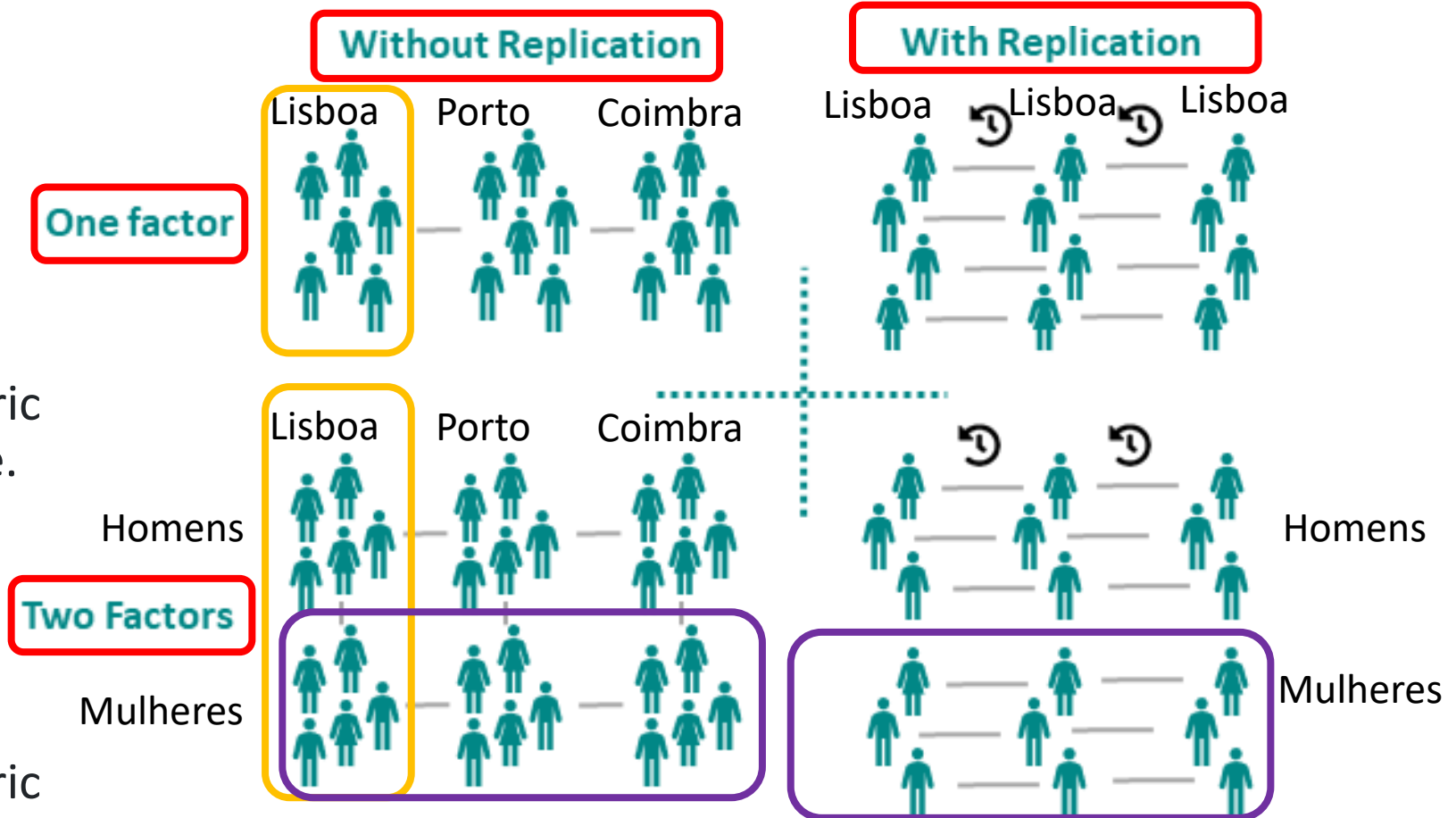
- Types:

- One-way ANOVA

- whether one independent variable has an influence on a metric dependent variable.

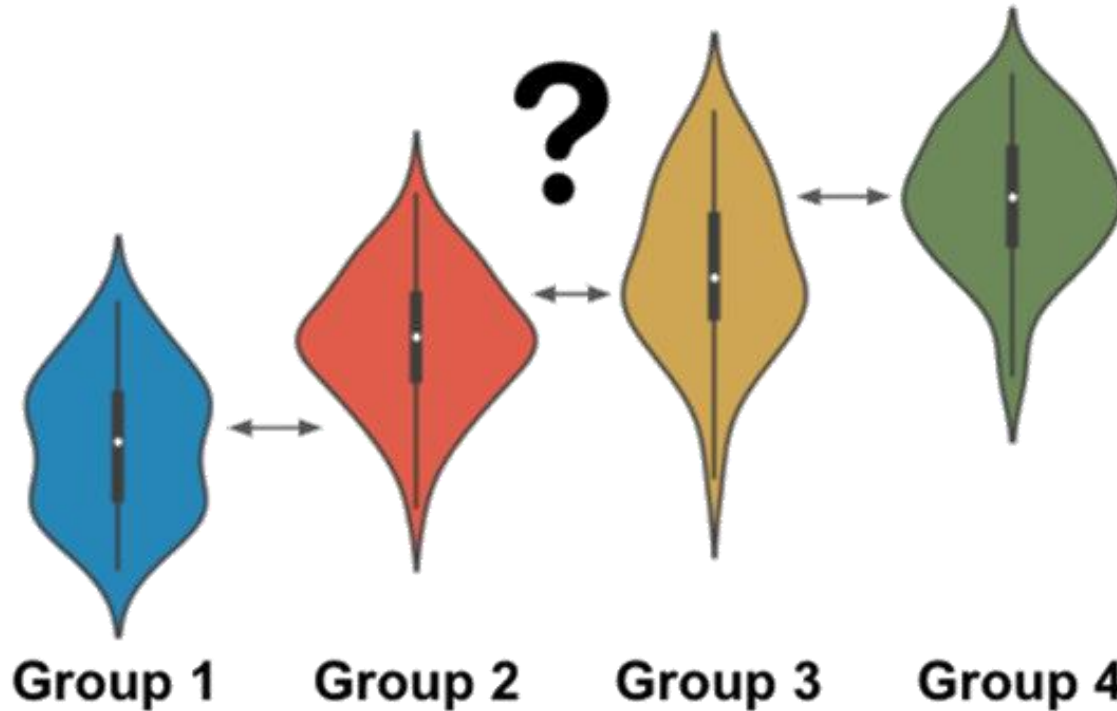
- Two-way ANOVA

- whether two independent variables have an influence on a metric dependent variable.



Inferential Statistics

- Parametric tests
- One-way ANOVA
 - sometimes called the One-Way ANOVA **F-Test**



<https://www.statstest.com/one-way-anova/>

- Null hypothesis H_0 : The mean value of all groups is the same.
- Alternative hypothesis H_1 : There are differences in the mean values of the groups.

<https://datatab.net/tutorial/one-sample-t-test>

Inferential Statistics

- Parametric tests
- One-way ANOVA
 - Requirements
 - Variables:
 - Independent variable: one nominal (with at least two values)
 - Dependent variable: one quantitative
 - **Homogeneity:** The variances in each group should be roughly the same. This can be checked with the Levene test.
 - **Normal distribution:** The data within the groups should be normally distributed. If this condition is not met, the non parametric Kruskal-Wallis test can be used.

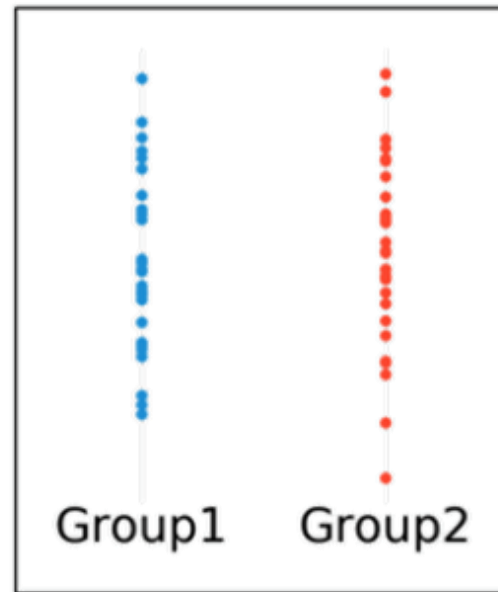
Inferential Statistics

- Parametric tests
- One-way ANOVA
 - Requirements
 - Homogeneity:
if the condition of variance
homogeneity is not fulfilled, **Welch's ANOVA** can be calculated instead of the "normal" ANOVA.

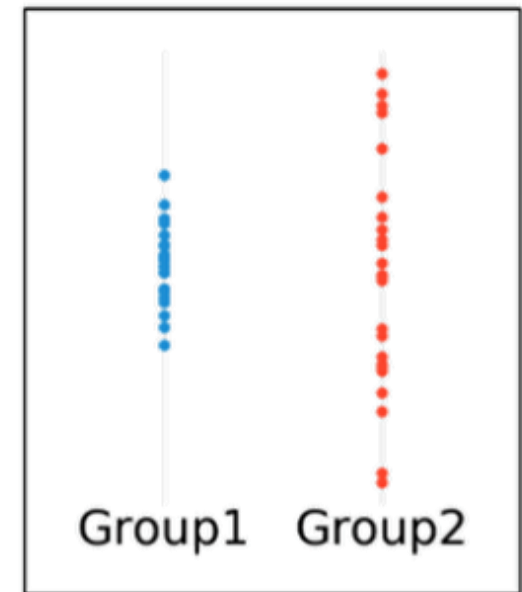
Teste de Levene:

H_0 = grupos possuem homogeneidade de variâncias:
 $\text{sig} > 0,05 \rightarrow$ aceitar $H_0 \rightarrow$ homogeneidade de variâncias

Homogeneity of Variance: making sure the variables take on reasonably similar values.



Group 1 and Group 2 have similar spread

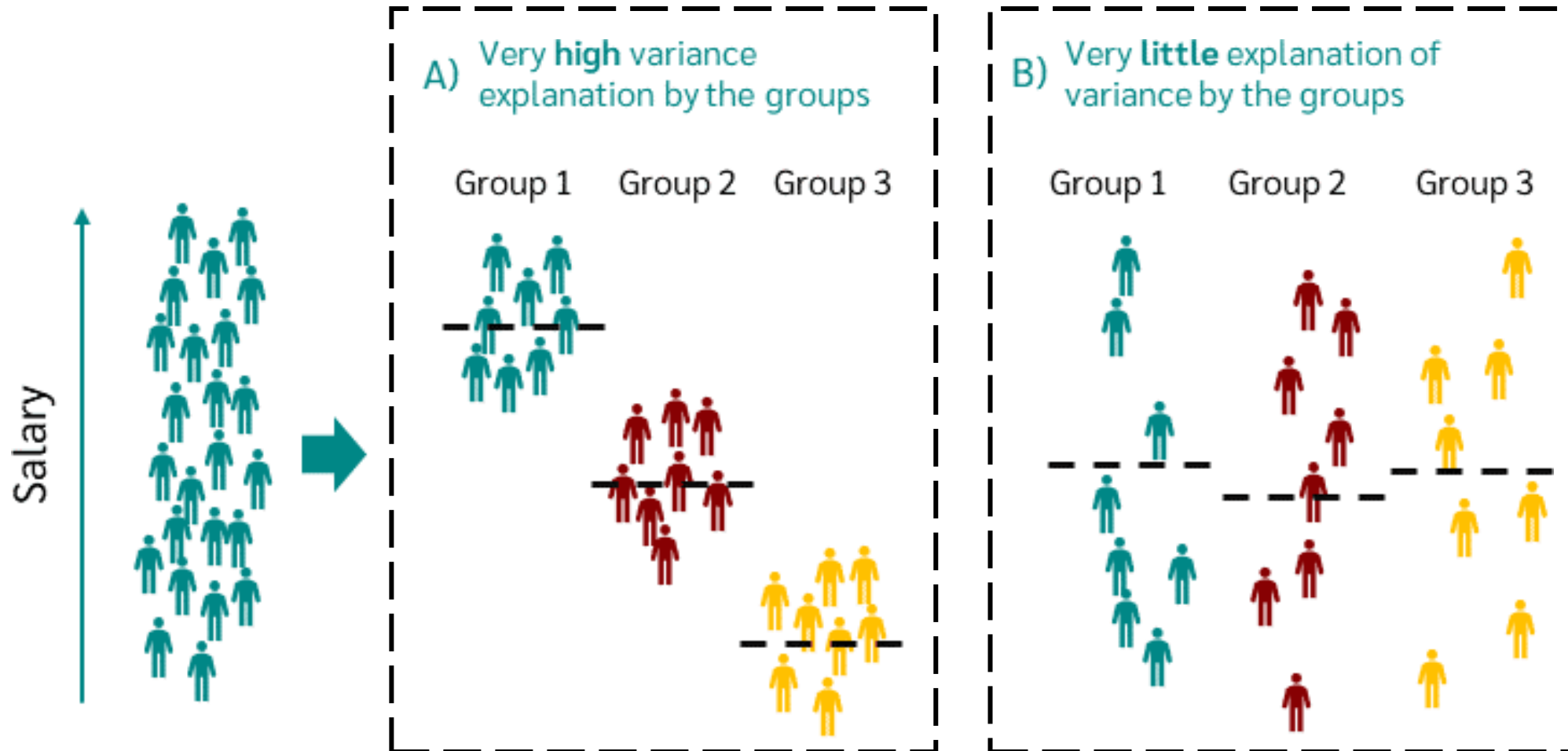


Group 2 is much more spread out than Group 1

Inferential Statistics

- Parametric tests
- One-way ANOVA

- A) the groups have a very high influence on the salary
B) the groups have very little influence on the salary



<https://datatab.net/tutorial/one-sample-t-test>

Inferential Statistics

- Parametric tests
- One-way ANOVA
 - Post-hoc Tests
 - also known as *multiple comparisons tests*
 - the results of the ANOVA can only make a statement about whether there are differences between at least two groups. However, it cannot be determined which groups are exactly different.
 - a **post-hoc test is needed to determine which groups differ.**

Inferential Statistics

- Parametric tests
- One-way ANOVA
 - Post-hoc Tests

Quando SE ASSUME que as variâncias entre os grupos são iguais:

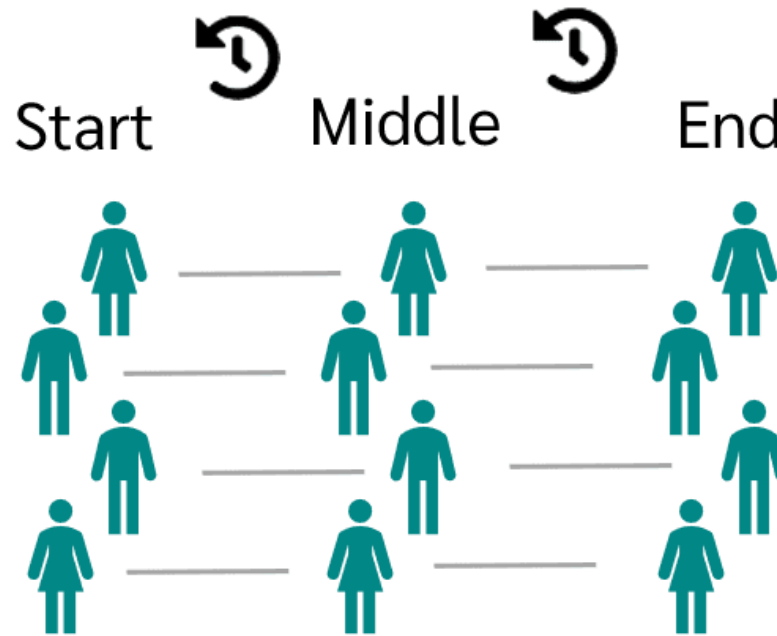
- teste Least Significant Difference de Fisher (LSD no SPSS)
 - teste Honestly Significant Difference de **Tukey** (Tukey no SPSS)
 - teste de **Scheffé**
 - teste de Newman Keuls (S-N-K no SPSS)
 - teste de Duncan
 - teste de **Gabriel**
 - teste de Hochberg's GT 2
 - teste de **Bonferroni**
 - teste R E G W Q
-
- para amostras grandes (>50)
- para amostras desequilibradas
- para amostras pequenas (<50)
mas não inferiores a 12

Quando NÃO SE ASSUME que as variâncias entre os grupos são iguais:

- teste de Dunnett's T 3
- teste de Dunnett's C
- teste de **Games-Howell**

Inferential Statistics

- Parametric tests
- One-way ANOVA with repeated measurements



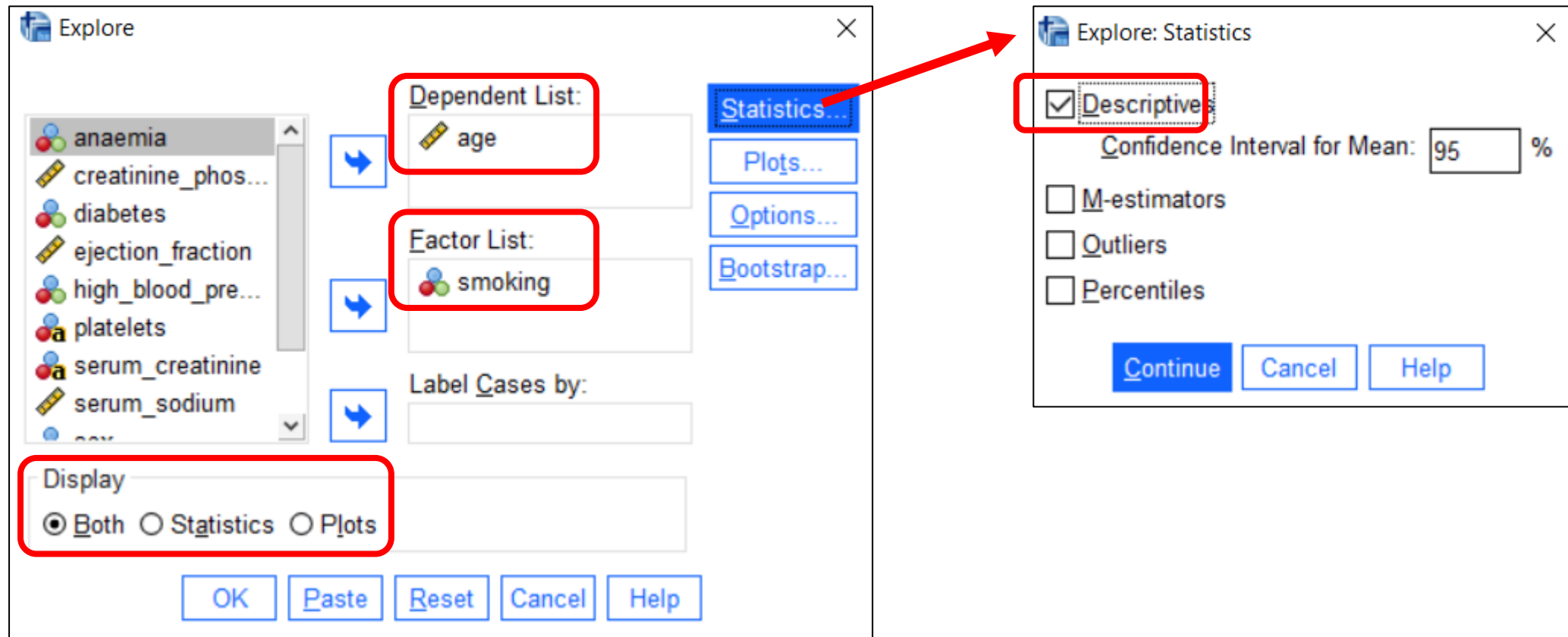
Does a two-week low-carb diet have an impact on body weight? To answer this question, the weight of 50 participants is measured at the beginning, in the middle and at the end

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- **SPSS**

- SPSS
- Normality Requirements

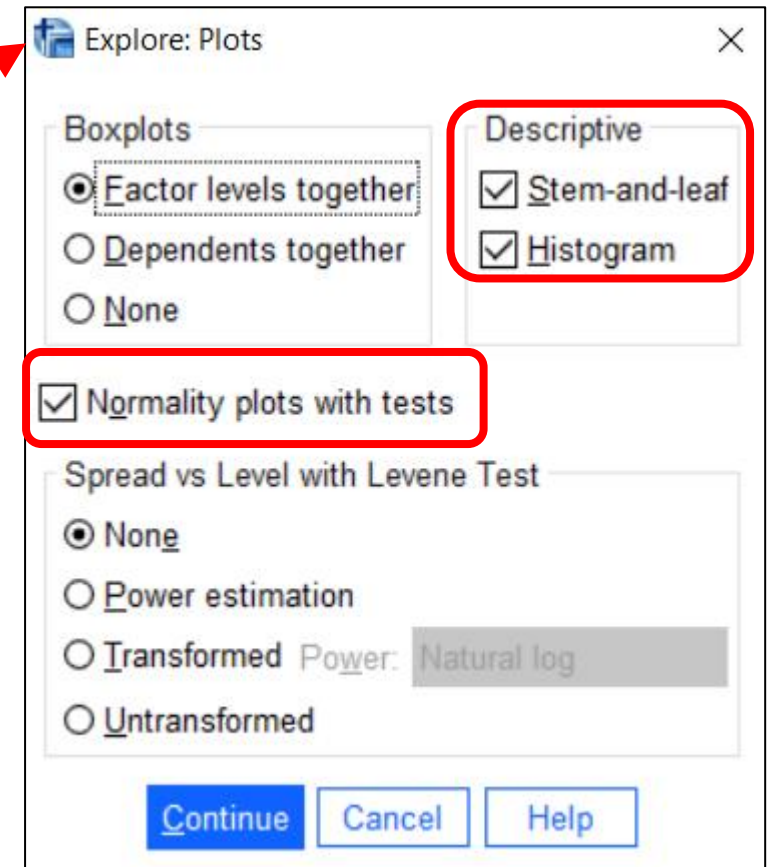
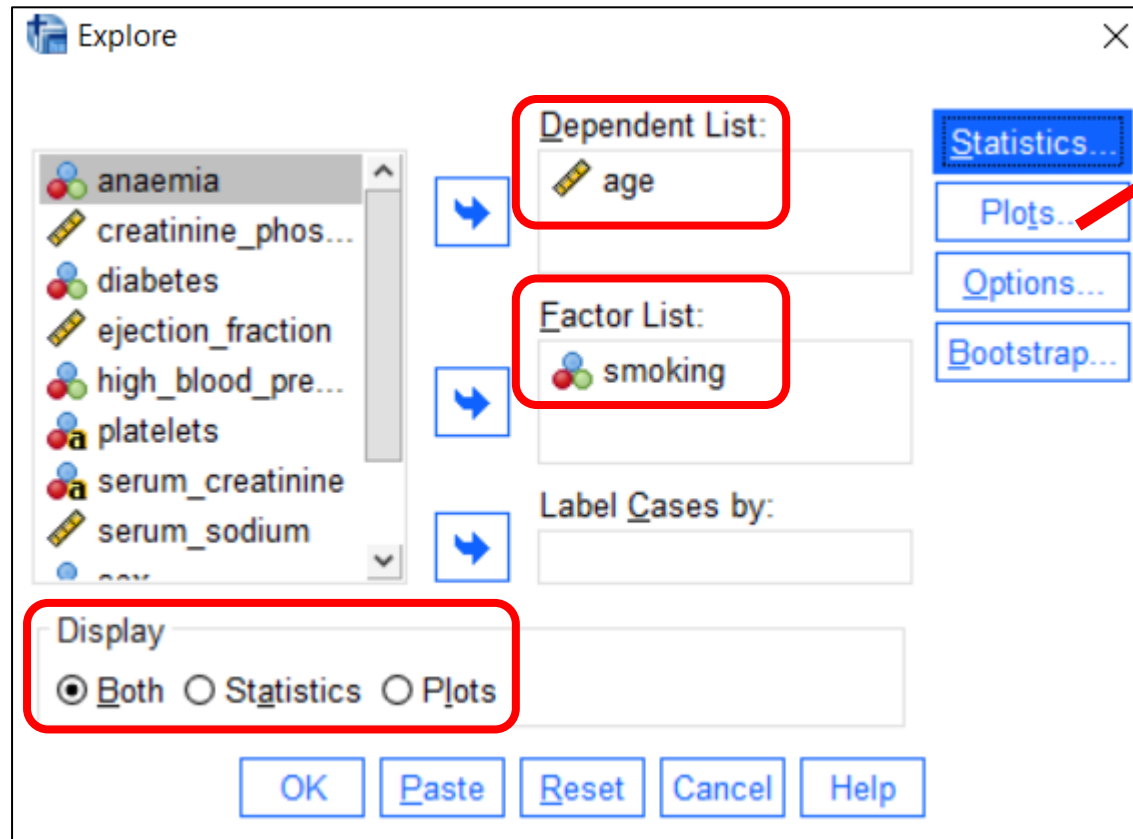
Analyse -> Descriptive Statistics -> Explore



Inferential Statistics

- **SPSS**
- **Normality Requirements**

Analyse -> Descriptive Statistics -> Explore



Inferential Statistics

- **SPSS**
- **Normality Requirements**
 - **Testes de Normalidade:**

Analyse -> Descriptive Statistics -> Explore

Case Processing Summary

		Valid		Cases Missing		Total	
		N	Percent	N	Percent	N	Percent
age	0	201	99,0%	2	1,0%	203	100,0%
	1	96	100,0%	0	0,0%	96	100,0%

> 50 casos -> **Kolmogorov–Smirnov (KS)**
< 50 casos -> **Shapiro–Wilk (SW)**

Tests of Normality

		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
age	0	,078	201	,005	,969	201	<,001
	1	,076	96	,200*	,981	96	,168

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

H0 = distribuição normal:
sig < 0,05 -> rejeitar H0 -> **sem distribuição normal**

H0 = distribuição normal:
sig > 0,05 -> aceitar H0 -> **distribuição normal**

Inferential Statistics

- SPSS
- Normality Requirements
 - Medidas de Assimetria:

Analyse -> Descriptive Statistics -> Explore

		Descriptives		
		smoking		
age	0		Statistic	Std. Error
		Mean	60,6816	,86182
		95% Confidence Interval for Mean		
		Lower Bound	58,9822	
		Upper Bound	62,3810	
		5% Trimmed Mean	60,1855	
		Median	60,0000	
		Variance	149,288	
		Std. Deviation	12,21835	
		Minimum	40,00	
		Maximum	95,00	
		Range	55,00	
		Interquartile Range	19,50	
		Skewness	,505	,172
		Kurtosis	-,095	,341

Distribuição simétrica =>

$-1,96 < \text{"coeficiente skewness"} / \text{"erro padrão skeness"} < 1,96$
considera-se assimetria tolerável se valor no intervalo $[-3; 3]$

Distribuição Mesocúrtica =>

$\text{"coeficiente kurtosis"} / \text{"erro padrão kurtosis"} \approx 3$

Skewness:

$0,505/0,172=2,93 < 3$

Distribuição simétrica

Kurtosis

$0,095/0,341=0,27 \neq 3$

Distribuição NÃO mesocúrtica

Inferential Statistics

- SPSS
- Normality Requirements
 - Medidas de Assimetria:

Analyse -> Descriptive Statistics -> Explore

Descriptives			Statistic	Std. Error
smoking	1	Mean	61,1563	1,16089
	95% Confidence Interval for Mean	Lower Bound	58,8516	
		Upper Bound	63,4609	
	5% Trimmed Mean		60,9676	
	Median		60,0000	
	Variance		129,375	
	Std. Deviation		11,37433	
	Minimum		40,00	
	Maximum		90,00	
	Range		50,00	
	Interquartile Range		18,00	
	Skewness		,220	,246
	Kurtosis		-,462	,488

Distribuição simétrica =>

$-1,96 < \text{"coeficiente skewness"} / \text{"erro padrão skewness"} < 1,96$
considera-se assimetria tolerável se valor no intervalo $[-3; 3]$

Distribuição Mesocúrtica =>

$\text{"coeficiente kurtosis"} / \text{"erro padrão kurtosis"} \approx 3$

Skewness:

$0,220/0,246=0,89 < 3$

Distribuição simétrica

Kurtosis

$0,462/0,488=0,95 \neq 3$

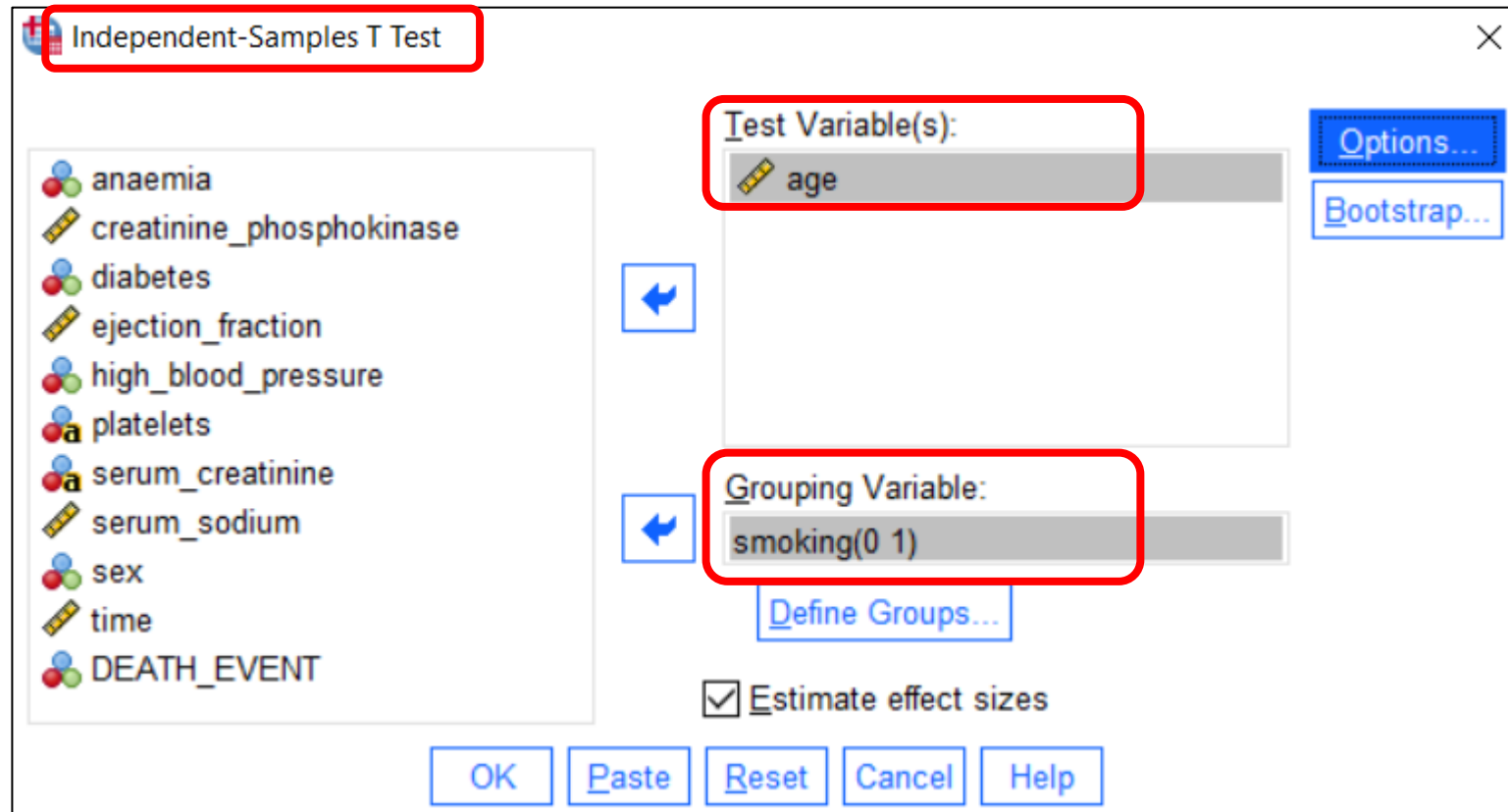
Distribuição NÃO mesocúrtica

Inferential Statistics

- SPSS
- Unpaired Samples t-test

Analyse -> Compare Means ->

- One-sample T teste
- Independent-samples T test
- Paired Samples T test



Inferential Statistics

- SPSS
- Unpaired Samples t-test

Analyse -> Compare Means ->

- One-sample T teste
- Independent-samples T test
- Paired Samples T test

Group Statistics

	smoking	N	Mean	Std. Deviation	Std. Error Mean
age	0	201	60,6816	12,21835	,86182
	1	96	61,1563	11,37433	1,16089

Independent Samples Test

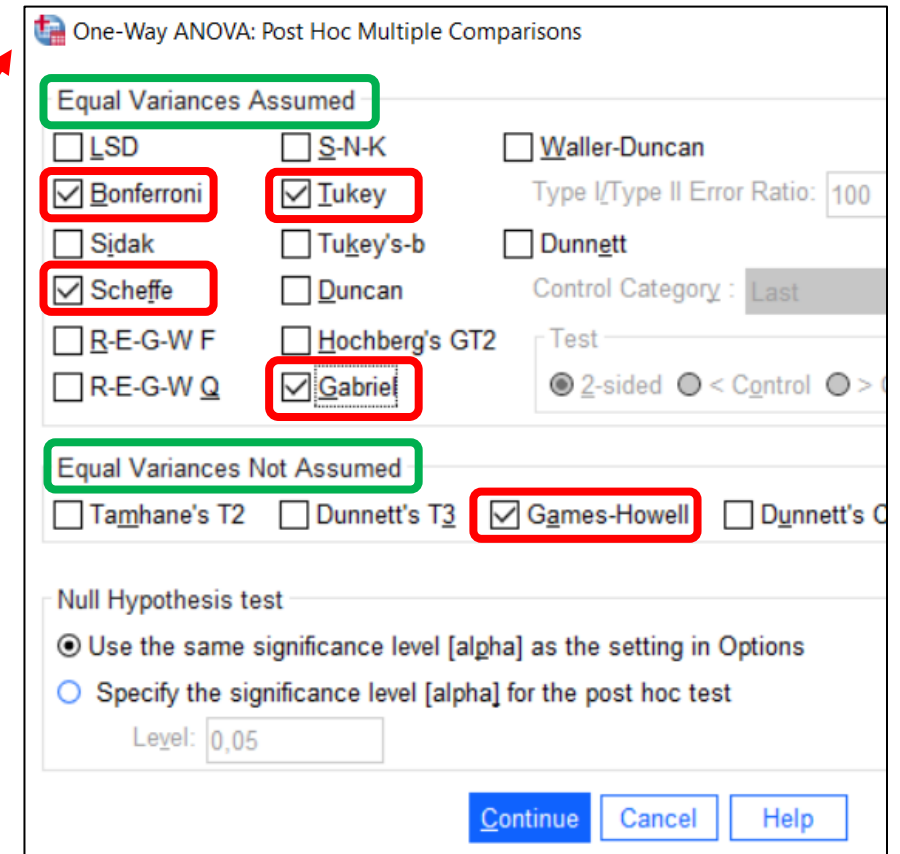
Levene's Test for Equality of Variances				t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
age	Equal variances assumed	,358	,550	-,320	295	,749	-,47466	1,48294	-3,39314	2,44382
	Equal variances not assumed			-,328	199,750	,743	-,47466	1,44582	-3,32568	2,37636

Teste de Levene: H_0 = grupos possuem homogeneidade de variâncias:
sig > 0,05 -> aceitar H_0 -> homogeneidade de variâncias

Teste t: H_0 = médias dos grupos são iguais:
sig > 0,05 -> aceitar H_0 -> médias dos grupos são iguais

- SPSS
- One-way ANOVA

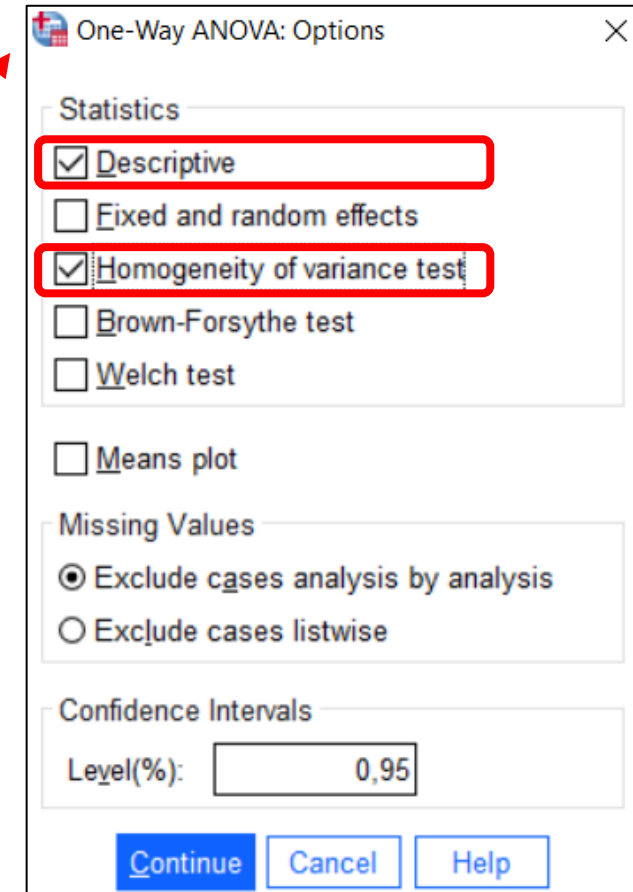
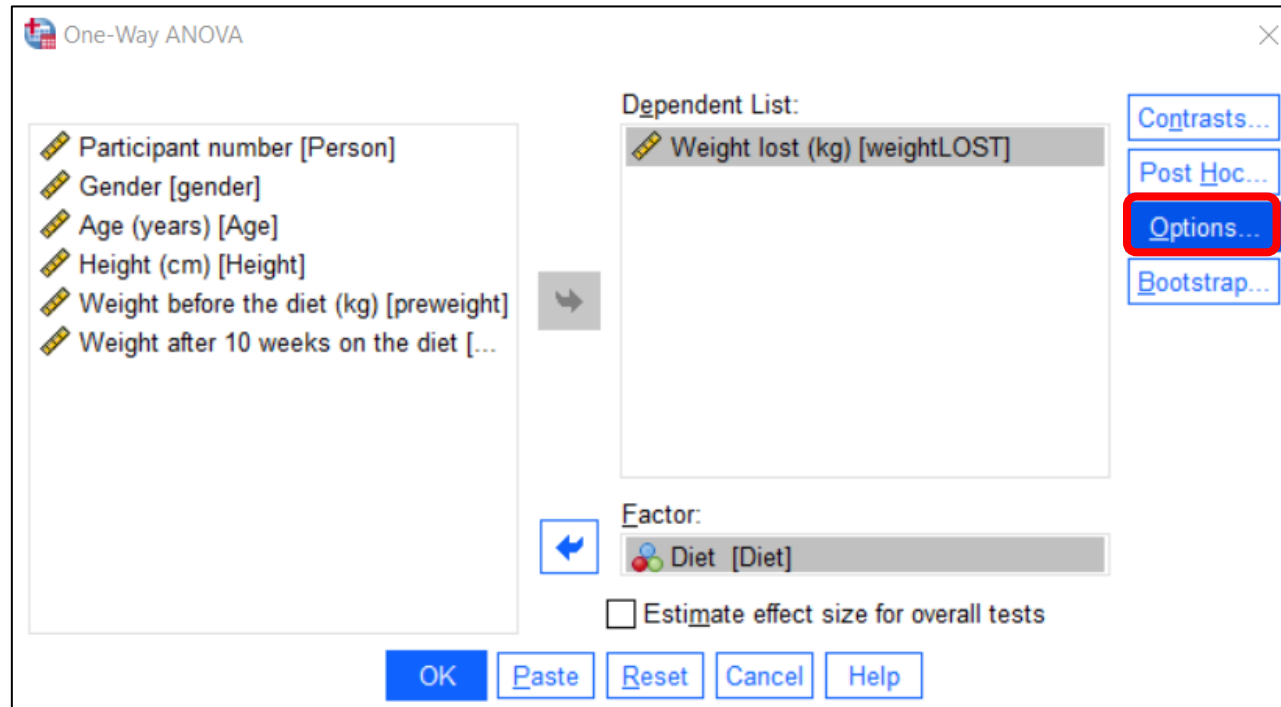
Analyse -> Compare Means -> One-Way ANOVA



Inferential Statistics

- SPSS
- One-way ANOVA

Analyse -> Compare Means -> One-Way ANOVA



Inferential Statistics

- **SPSS**
- **One-way ANOVA**

Analyse -> Compare Means -> **One-Way ANOVA**

Descriptives

Weight lost (kg)									
		N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
						Lower Bound	Upper Bound		
1	24	3,3000	2,24015	,45727	2,3541	4,2459	-,60	9,00	
2	27	3,0259	2,52337	,48562	2,0277	4,0241	-2,10	7,90	
3	27	5,1481	2,39557	,46103	4,2005	6,0958	,50	9,20	
Total	78	3,8449	2,55148	,28890	3,2696	4,4201	-2,10	9,20	

Inferential Statistics

- SPSS
- One-way ANOVA

Analyse -> Compare Means -> One-Way ANOVA

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
Weight lost (kg)	Based on Mean	,659	2	75	,520
	Based on Median	,626	2	75	,538
	Based on Median and with adjusted df	,626	2	73,794	,538
	Based on trimmed mean	,675	2	75	,512

Teste de Levene:

H_0 = grupos possuem homogeneidade de variâncias:
sig > 0,05 -> aceitar H_0 -> homogeneidade de variâncias

ANOVA

Weight lost (kg)

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	71,094	2	35,547	6,197	,003
Within Groups	430,179	75	5,736		
Total	501,273	77			

ANOVA:

H_0 = médias dos grupos são iguais:
sig < 0,05 -> rejeitar H_0 -> médias dos grupos são diferentes

ANOVA apenas permite concluir que existem pelo menos duas médias diferentes, mas não nos diz quais. Para tal será necessário a análise do **quadro das comparações múltiplas** (*testes post-hoc*).

Inferential Statistics

- SPSS
- One-way ANOVA

Analyse -> Compare Means -> One-Way ANOVA

Multiple Comparisons

Dependent Variable: Weight lost (kg)

	(I) Diet	(J) Diet	Mean Difference (I- J)	Std. Error	Sig.	95% Confidence Interval	
Bonferroni para amostras pequenas (<50)	1	2	,27407	,67188	1,000	-1,3713	1,9194
		3	-1,84815*	,67188	<u>,022</u>	-3,4935	-,2028
	2	1	-,27407	,67188	1,000	-1,9194	1,3713
		3	-2,12222*	,65182	<u>,005</u>	-3,7184	-,5260
	3	1	1,84815*	,67188	,022	,2028	3,4935
		2	2,12222*	,65182	,005	,5260	3,7184

Testes Post Hoc:

H_0 = médias dos grupos são iguais:
sig < 0,05 -> rejeitar H_0 -> médias dos grupos
são diferentes

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